

RACIBORZ, Poland

WMO: 125400

Lat: 50.0610N

Lon: 18.1908E

Elev: 206

StdP: 98.87

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-11.7	-17.3	0.8	-14.5	-14.3	1.1	-10.9	11.7	2.3	10.7	2.3	2.0	150	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	30.5	20.2	28.5	19.5	26.8	18.8	21.3	27.7	20.4	26.5	19.6	25.2	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.3	23.9	18.4	13.6	22.9	17.6	12.9	22.1	62.7	27.7	59.5	26.5	56.8	25.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.5	7.4	-17.4	33.6	4.5	1.7	-20.6	34.8	-23.3	35.8	-25.8	36.8	-29.1	38.0		
			-18.1	23.3	4.5	1.0	-21.3	23.9	-23.9	24.5	-26.5	25.0	-29.8	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-1.6	0.0	3.9	9.7	13.9	17.3	19.2	19.0	14.4	9.9	5.3	0.6
	DBStd	8.32	5.28	5.08	4.46	4.07	3.65	3.43	3.16	3.24	3.41	4.04	4.05	4.83
	HDD10.0	1387	359	279	192	54	8	0	0	0	4	54	147	291
	HDD18.3	3424	617	512	447	261	143	61	27	30	124	262	391	549
	CDD10.0	1152	0	0	3	44	129	219	286	279	136	50	6	0
	CDD18.3	147	0	0	0	1	6	29	55	51	6	1	0	0
	CDH23.3	1479	0	0	0	16	81	280	550	485	66	2	0	0
	CDH26.7	411	0	0	0	1	10	69	170	150	11	0	0	0
Wind	WSAvg	3.2	3.8	3.7	3.8	3.4	3.1	2.9	2.8	2.6	2.8	3.0	3.3	3.8
Precipitation	PrecAvg	629	32	27	35	41	73	80	94	68	61	44	42	32
	PrecMax	914	61	68	78	77	233	147	314	143	134	103	91	81
	PrecMin	348	9	4	12	8	17	16	11	12	15	5	0	11
	PrecStd	114	13	14	19	19	41	33	65	34	33	27	21	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	12.3	18.1	25.4	27.9	31.4	32.7	33.2	28.3	22.9	17.2	12.1
		MCWB	8.3	8.6	10.9	15.7	17.8	21.2	20.6	20.4	18.1	16.2	12.8	9.3
	2%	DB	8.2	9.9	15.0	22.0	25.6	28.6	30.5	30.3	25.1	20.4	14.6	9.7
		MCWB	6.4	7.1	9.8	13.9	17.2	20.1	20.1	20.0	17.4	15.2	11.3	7.9
	5%	DB	6.5	8.2	12.6	19.5	23.5	26.5	28.5	28.0	22.9	18.2	12.8	7.9
		MCWB	5.0	5.9	8.6	12.5	16.5	18.9	19.6	19.5	16.6	14.1	10.2	6.3
	10%	DB	5.0	6.4	10.4	17.2	21.4	24.4	26.5	26.0	20.9	16.1	11.1	6.3
		MCWB	3.7	4.6	7.4	11.4	15.3	18.1	18.9	18.6	15.8	13.1	9.2	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	8.9	11.7	16.4	19.6	22.3	22.5	22.5	19.4	17.1	13.0	9.6
		MCDB	10.1	12.0	16.8	23.4	25.8	29.1	30.0	28.6	25.6	21.5	16.5	11.7
	2%	WB	6.6	7.3	10.2	14.7	18.1	21.0	21.3	20.9	18.2	15.6	11.6	7.9
		MCDB	8.1	9.6	14.3	21.2	23.6	27.2	27.6	27.8	22.8	19.2	14.3	9.7
	5%	WB	5.1	6.0	8.9	13.2	17.0	19.8	20.4	20.1	17.3	14.5	10.4	6.4
		MCDB	6.4	7.9	12.0	18.3	22.4	24.9	26.4	26.4	21.7	17.6	12.4	7.8
	10%	WB	3.8	4.7	7.5	11.9	15.9	18.7	19.6	19.2	16.3	13.3	9.3	5.1
		MCDB	4.9	6.3	10.2	16.3	20.5	23.2	25.0	24.8	20.2	16.0	11.0	6.3
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.8	7.5	10.2	10.2	10.1	10.9	10.9	9.3	8.0	5.6	4.7
		MCDBR	5.4	8.1	12.1	14.4	14.2	14.0	14.9	14.6	13.4	11.5	8.4	6.2
		MCWBR	4.4	5.8	7.5	7.5	7.0	6.7	5.9	5.8	6.5	6.9	5.9	4.9
	5% WB	MCDBR	5.3	7.7	10.9	13.1	12.7	12.3	13.0	12.8	11.6	10.4	7.9	6.0
		MCWBR	4.3	5.8	7.1	7.2	6.7	6.5	5.8	5.6	6.0	6.5	5.7	4.8
Clear-Sky Solar Irradiance	taub	0.316	0.341	0.381	0.434	0.429	0.429	0.444	0.435	0.404	0.373	0.346	0.312	
	taud	2.306	2.280	2.230	2.133	2.191	2.230	2.195	2.229	2.286	2.342	2.362	2.363	
	Ebn at Noon	710	784	813	806	830	832	811	798	782	738	668	662	
	Edh at Noon	69	91	114	140	138	134	137	126	108	86	67	58	
All-Sky Solar Radiation	RadAvg	0.84	1.59	2.68	4.08	4.92	5.23	5.02	4.52	3.21	1.91	0.94	0.65	
	RadStd	0.15	0.25	0.43	0.51	0.61	0.52	0.60	0.40	0.47	0.28	0.13	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	+0.50	N/A	N/A	+0.75	+0.37	+0.33	N/A	-154	+63	+22

Nomenclature: See separate page